Optimization-Induced Dynamics and the Inverted-U: A Distillation-Based Theory of Optimal Misalignment

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Abstract

Conventional approaches to AI alignment aim to minimize divergence between systems, implicitly assuming that perfect agreement maximizes utility. In this work, we challenge this assumption by proposing Ma Theory, which posits an inverted-U relationship between structural misalignment (ε^*) and emergent utility (E).

We introduce a Dual-Track Design to isolate the role of generative dynamics: (A) a distillation track based on stochastic gradient descent (SGD) trajectories across Pythia model checkpoints (step 0–128,000), and (B) an interpolation track using linear weight interpolation in LoRA parameter space. Experiments follow a pre-registered protocol with 50 samples across 3 seeds evaluated on WikiText-2.

We find that the distillation track exhibits a statistically significant inverted-U relationship in Pythia-410m (post-peak decline: $r = -0.993$, $p = 0.0007$), with a peak at $\varepsilon^* \approx 68$. The unimodal structure is replicated in Pythia-160m under identical experimental conditions, with a peak at $\varepsilon^* \approx 35$ (step 32,000), demonstrating that the phenomenon is replicated across model scales. In contrast, the interpolation track shows monotonic increase with no evidence of a peak.

We show that this difference is governed by the dynamics of novelty (D_{nov}): distillation induces early saturation ($D_{\text{nov}} \approx 0.98$), while interpolation produces approximately linear growth without saturation. This saturation is a key enabling condition for the emergence of the inverted-U.

While multiple asymmetric functions can fit the observed curve, the theoretically derived form $E = C\varepsilon^* \exp(-k\varepsilon^*)$ provides an interpretable minimal model, whereas log-normal variants offer higher descriptive flexibility under certain fitting conditions.

We conclude that the inverted-U is not observed along linear interpolation paths in parameter space, but appears only under optimization-induced dynamics such as SGD. The experimental evidence is consistent with the choice of generative dynamics playing a causal role in the formation of the inverted-U structure.

1 Introduction

Recent AI alignment and model integration methods [12, 7] have treated the minimization of distributional divergence—specifically, the KL divergence (ε^*) between teacher and student models—as the ideal objective. However, the Ma Theory proposed in this paper identifies a *translation paradox*: pursuing "perfect agreement" eliminates emergent utility, which arises from the productive interaction between systems. We call this framework *Ma Theory*, from the concept of *ma* ('in-between')—the productive space between cognitive systems from which emergence arises. This study demonstrates the existence of an inverted-U relationship between structural divergence (ε^*) and emergent utility (E) [13, 15], where emergence is maximized at an optimal degree of imperfection.

The central contribution of this paper is the mathematical and experimental demonstration that this inverted-U relationship is not observed under linear weight interpolation in parameter space, but appears only through the optimization-induced exploration trajectory of stochastic gradient descent (SGD) in the distillation process. Weight interpolation corresponds to a

linear path in parameter space, while distillation corresponds to a qualitatively different exploration trajectory driven by SGD. This contrast enables us to test whether the emergence of the inverted-U depends on the geometric properties of the learning path. We further replicate the phenomenon across two model scales (Pythia-410m and Pythia-160m) to assess its generality. We adopt a preregistered protocol, rigorously evaluating the existence of the inverted-U (H1) against the null hypothesis of monotonic increase (H0), as well as the dynamical limitations of the generative process (H2.5), under a falsifiable experimental design.

2 Theory

This section defines the fundamental theorems constituting Ma Theory and the mathematical model governing the dynamics of emergent utility.

We define the structural divergence ε^* between systems A and B as the KL divergence [8] between their output probability distributions:

$$\varepsilon^* = D_{\mathrm{KL}}(p_A \| p_B)$$

The emergent utility E (specifically, agreement-based emergence E_{agree}) is decomposed into the product of embedding-based novelty D_{nov} and output token agreement probability P_{agree}:

$$E = D_{\mathrm{nov}} \times P_{\mathrm{agree}}$$

The embedding vector ϕ and novelty D_{nov} are defined as:

$$\phi(x;\theta) = \mathrm{L2Norm}\left(\frac{1}{T}\sum_{t=1}^{T} h_t^{(L)}(x;\theta)\right), \quad D_{\mathrm{nov}} = 1 - \phi_A \cdot \phi_B$$

where $h_t^{(L)}$ is the hidden state vector at token t in the final hidden layer, and L2Norm denotes normalization by the Euclidean norm. The agreement probability P_{agree} is defined as:

$$P_{\mathrm{agree}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\left[\arg\max_v p_A(v|x_{<t}) = \arg\max_v p_B(v|x_{<t})\right]$$

Theorem 1 (Complete Translation Impossibility)**.** *When cognitive systems A and B are distinct ($A \neq B$), the structural divergence is always positive: $\varepsilon^* > 0$.*

Proof. By assumption, $A \neq B$ implies $p_A \neq p_B$, i.e., there exists at least one context $x_{<t}$ and token v such that $p_A(v|x_{<t}) \neq p_B(v|x_{<t})$. By Gibbs' inequality, $D_{\mathrm{KL}}(p_A\|p_B) \geq 0$ with equality if and only if $p_A = p_B$. Since $p_A \neq p_B$, we obtain $\varepsilon^* = D_{\mathrm{KL}}(p_A\|p_B) > 0$. □

Proposition 1 (Optimal Imperfection)**.** *Under distillation dynamics, the emergent utility E forms an inverted-U as a function of ε^*. The generalized functional form is:*

$$E(\varepsilon^*) = C\,(\varepsilon^*)^{\alpha}\,\exp(-k\varepsilon^*)$$

where $C > 0$ is a scaling constant, $k > 0$ governs the decay rate, and $\alpha > 0$ controls the rise exponent. As the theoretical baseline, we fix $\alpha = 1$, yielding the minimal two-parameter form $E = C\varepsilon^* \exp(-k\varepsilon^*)$ with optimal divergence $\varepsilon^*_{\mathrm{opt}} = 1/k$.

Note: This proposition provides a theoretically grounded minimal model derived from rate-distortion theory [4], whose parameters (C, k) admit physical interpretation and which yields the optimal divergence $\varepsilon^*_{\mathrm{opt}} = 1/k$ as a direct prediction. In data-driven fitting, a log-normal functional form (LogPeak) may yield a lower AIC in some conditions. However, regardless of

the functional form employed, the inverted-U structure decisively rejects quadratic and linear models. Experimentally, α is not stable across model scales ($\alpha \approx 0.81$ for 410m, $\alpha \approx 1.62$ for 160m under unweighted fitting), and the AIC ranking of specific functional forms depends on both the fitting method and model scale. Proposition 1 should therefore be regarded as a theoretically grounded minimal model rather than a universally optimal functional form.

Theorem 2 (Non-Understanding as Source of Emergence)**.** *Under the assumption that the embedding map ϕ is injective with respect to the output distribution p (i.e., $\phi_A \neq \phi_B$ implies $p_A \neq p_B$), a necessary condition for the observation of positive emergence is the existence of incomplete understanding (divergence) between systems: $E > 0 \implies \varepsilon^* > 0$.*

Proof. If $E = D_{\text{nov}} \times P_{\text{agree}} > 0$, then both factors are positive. $D_{\text{nov}} > 0$ implies $\phi_A \cdot \phi_B < 1$, hence $\phi_A \neq \phi_B$. By the injectivity assumption on ϕ, distinct embeddings imply distinct output distributions: $p_A \neq p_B$. By Theorem 1, $\varepsilon^* > 0$.

Remark on the injectivity assumption. In the architectures considered (Pythia/GPT-NeoX), the output distribution is computed as $p = \text{softmax}(W_{\text{unembed}}\, h^{(L)})$, and the embedding ϕ is a deterministic function of $h^{(L)}$ involving token-wise averaging and L2 normalization. The natural direction of dependence is $h^{(L)} \to p$ and $h^{(L)} \to \phi$. The injectivity assumption ($\phi_A \neq \phi_B \Rightarrow p_A \neq p_B$) is adopted as an architecture-dependent working assumption in this paper. Because ϕ involves averaging over token positions, information may in principle be lost, and a full-rank unembedding matrix alone does not guarantee injectivity of the composed map. Empirical verification or the identification of stronger sufficient conditions is left to future work. □

Dynamical Derivation of the Inverted-U (The Friction Update). The conditions for the emergence of the inverted-U described in Proposition 1 are mathematically explained by the following differential equation. We model the discrete checkpoints as samples from a continuous, differentiable learning trajectory parameterized by ε^*:

$$\frac{dE}{d\varepsilon^*} = \frac{dD_{\text{nov}}}{d\varepsilon^*} \times P_{\text{agree}} + D_{\text{nov}} \times \frac{dP_{\text{agree}}}{d\varepsilon^*}$$

In distillation dynamics (SGD trajectory), the novelty D_{nov} saturates at its upper bound (≈ 0.98) during early learning, yielding $\frac{dD_{\text{nov}}}{d\varepsilon^*} \approx 0$. Consequently, the second term (negative contribution) becomes dominant, forming the right-side decay region of the inverted-U.

In contrast, under weight interpolation (linear path in parameter space), $D_{\text{nov}} \propto \varepsilon^*$ (linear increase) without saturation. The positive contribution of the first term consistently dominates, resulting in $\frac{dE}{d\varepsilon^*} > 0$ (perpetual monotonic increase). Therefore, the inverted-U is not expected to form along linear interpolation paths in parameter space, and is not observed in our experiments.

3 Methods

3.1 Dual-Track Experimental Design

This study adopts a "Dual-Track Design" that examines the relationship between structural divergence ε^* and emergent utility E under two fundamentally different generative processes.

Track A: Distillation measures the divergence ε^* between each checkpoint and the final model as a single neural network progresses from initialization to its final state through pretraining. This trajectory is an optimization-induced exploration process driven by SGD [10].

Track B: Interpolation controls ε^* by linearly interpolating the weights of a base model and a LoRA fine-tuned model, varying the interpolation coefficient $\lambda \in [0, 1]$. Linear movement in parameter space constitutes a Euclidean geodesic in parameter space [2, 9, 16], though it does

not in general correspond to a geodesic on the statistical manifold under Fisher information geometry [3].

This contrast tests whether the emergence of the inverted-U depends not only on the magnitude of ε^* but on the geometric properties of the path that generates it.

3.2 Models

All experiments use EleutherAI/Pythia models [5], autoregressive language models based on the GPT-NeoX architecture [6]. All pretraining checkpoints (step 0–143,000) are publicly available on HuggingFace.

Primary scale (410m): For Track A, step 143,000 serves as the teacher model, with steps {0, 1000, 2000, 4000, 8000, 16000, 32000, 64000, 128000} as student models. Step 0 corresponds to the fully initialized state (random weights). For Track B, a student model was created by applying LoRA fine-tuning [11] (rank=64, α=256, target modules: query, key, value) to the Pythia-410m base model. The interpolated model weights are constructed as $W(\lambda) = W_{\text{base}} + \lambda \cdot \Delta W$, with 16 values of λ measured.

LoRA fine-tuning was conducted on `code_search_net/python` (train split, first 5,000 samples) with a constant learning rate of 1×10^{-4} (no scheduler), a custom training loop (manual backward pass), for 10,000 steps. A preliminary verification confirmed that `merge_and_unload()` has negligible effect on ε^* (merge/LoRA ratio = 0.9997).

Replication scale (160m): To assess scale generality, the distillation track was replicated using Pythia-160m with identical protocol: teacher at step 143,000, the same 9 student checkpoints, and identical evaluation conditions (WikiText-2 test, 50 samples $\times$ 3 seeds).

3.3 Measurement Protocol

All experiments use a unified measurement codebase. Evaluation data consists of sequences extracted from the test split of WikiText-2. The following four metrics are measured:

Structural divergence ε^*:

$$\varepsilon^* = \sum_{t=1}^{T} D_{\text{KL}}(p_T(\cdot|x_{<t}) \| p_S(\cdot|x_{<t}))$$

Novelty D_{nov}:

$$\phi(x;\theta) = \text{L2Norm}\left(\frac{1}{T}\sum_{t=1}^{T} h_t^{(L)}(x;\theta)\right), \quad D_{\text{nov}} = 1 - \phi_T \cdot \phi_S$$

Agreement probability P_{agree}:

$$P_{\text{agree}} = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\left[\arg\max_v p_T(v|x_{<t}) = \arg\max_v p_S(v|x_{<t})\right]$$

Emergent utility E_{agree}:

$$E_{\text{agree}} = D_{\text{nov}} \times P_{\text{agree}}$$

All four metrics are computed per input sequence x. The reported values are sample means over $N = 50$ sequences drawn from the evaluation dataset $\mathcal{D}$:

$$\bar{E}_{\text{agree}} = \frac{1}{N}\sum_{i=1}^{N} E_{\text{agree}}(x_i), \quad x_i \sim \mathcal{D}$$

where $E_{\text{agree}}(x_i) = D_{\text{nov}}(x_i) \times P_{\text{agree}}(x_i)$ is computed independently for each sequence. This procedure is repeated for each of 3 seeds (controlling sequence sampling), and the final estimates are seed-averaged with standard errors.

Each condition is measured with 50 samples $\times$ 3 seeds, reporting means and standard errors. KL divergence is computed in float32 precision. Each input sequence is truncated to a maximum of 512 tokens without padding; sequences shorter than 512 tokens retain their natural length. Since ε^* is defined as a sum over tokens, it depends on the effective sequence length T. However, because all metrics are computed per sequence and then averaged across sequences, the per-sample variation in T is absorbed into the sample mean, and cross-condition comparisons (which use identical input sequences) are not affected.

3.4 Pre-registration and Hypothesis Testing

To eliminate cherry-picking, the following hypothesis testing structure was preregistered:

- **H0 (Null):** E_{agree} increases monotonically with ε^*; no inverted-U exists.
- **H1a:** E_{agree} exhibits an inverted-U with an interior peak $\varepsilon^*_{\text{opt}}$.
- **H1b:** The functional form of Proposition 1 provides the best fit among candidate models.
- **H2.5:** The emergence of the inverted-U depends on the generative process (distillation vs. interpolation).

Rejection of H0 uses a Pearson correlation significance test on the post-peak decay, with Welch's t-test for adjacent-point comparisons.

3.5 Model Comparison

To evaluate the functional form of Proposition 1, five candidate models are compared using the Akaike Information Criterion (AIC) [1]:

1. Prop. 1 original (2p): $E = C\varepsilon^* \exp(-k\varepsilon^*)$
2. Prop. 1 gamma (3p): $E = C(\varepsilon^*)^\alpha \exp(-k\varepsilon^*)$
3. Log-peak (3p): $E = C \exp\left(-\frac{(\ln \varepsilon^* - \mu)^2}{2\sigma^2}\right)$
4. Quadratic (3p): $E = a(\varepsilon^*)^2 + b\varepsilon^* + c$
5. Linear (2p): $E = a\varepsilon^* + b$

Model comparison is conducted under both weighted least squares (inverse standard error weighting) and unweighted least squares to verify the stability of model rankings across fitting methods.

4 Results

4.1 Distillation Track: Confirmation of the Inverted-U (Pythia-410m)

Figure 1 and Table 1 present all measurements from the distillation track. E_{agree} exhibits a clear inverted-U with respect to ε^*, peaking at step 8,000 ($\varepsilon^* = 67.57$, $E_{\text{agree}} = 0.468$).

Statistical verification: The correlation between E_{agree} and ε^* in the post-peak region is $r = -0.993$, $p = 0.0007$, confirming a highly significant decay trend. Adjacent-point differences near the peak do not reach significance (peak vs. step 16,000: $t = 0.843$, $p = 0.20$; peak vs. step 4,000: $t = 0.394$, $p = 0.35$), consistent with a broad plateau in the range $\varepsilon^* \approx 50$–90.

The null hypothesis H0 (monotonic increase) is rejected at $p = 0.0007$.

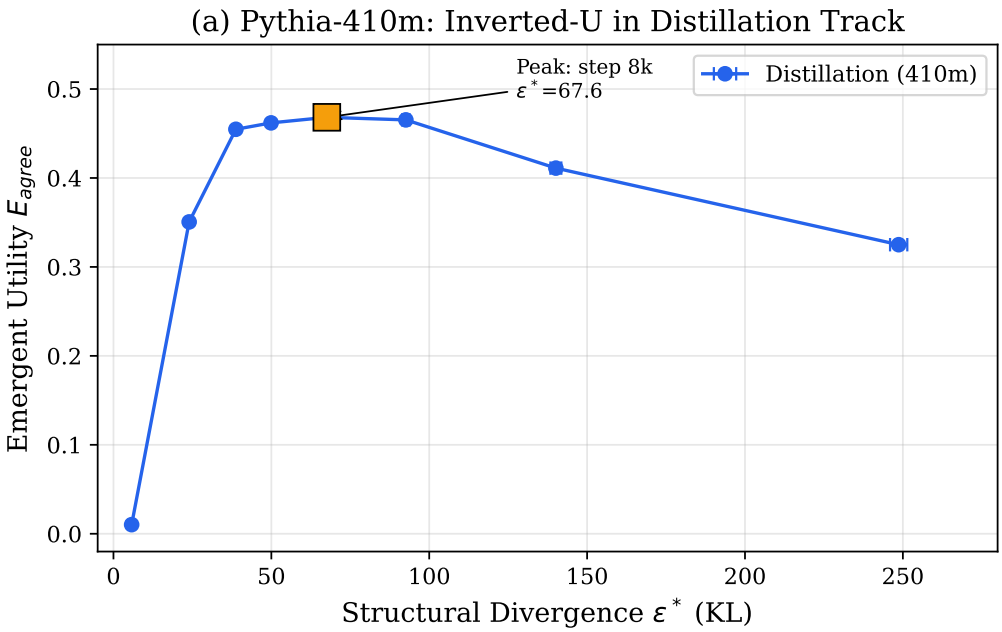

Figure 1: Pythia-410m distillation track: E_{agree} vs. ε^*. The inverted-U peaks at step 8,000 ($\varepsilon^* = 67.57$). Step 0 ($\varepsilon^* = 969.9$, $E_{\text{agree}} = 0$) is omitted for readability; see Table 1. Error bars indicate standard errors of ε^* across 3 seeds.

Table 1: Distillation track (Pythia-410m): All measurements. $\star$ marks the peak of E_{agree}.

Step	ε^* (mean $\pm$ SE)	D_{nov}	P_{agree}	E_{agree}
0	969.90 ± 4.01	0.982	0.000	0.000
1,000	248.67 ± 2.73	0.886	0.366	0.325
2,000	140.11 ± 1.73	0.828	0.497	0.411
4,000	92.60 ± 1.16	0.815	0.571	0.465
8,000	67.57 ± 0.86	0.745	0.628	$0.468^\star$
16,000	49.91 ± 0.68	0.684	0.674	0.462
32,000	38.77 ± 0.56	0.638	0.713	0.455
64,000	23.97 ± 0.48	0.458	0.765	0.351
128,000	5.76 ± 0.09	0.012	0.874	0.010

4.2 Scaling Replication (Pythia-160m)

To assess whether the inverted-U structure generalizes beyond a single model scale, the distillation track was replicated using Pythia-160m under identical experimental conditions (Figure 2 and Table 2).

The inverted-U structure is replicated: E_{agree} peaks at step 32,000 ($\varepsilon^* = 34.96$, $E_{\text{agree}} = 0.644$), with a post-peak monotonic decline observed across the subsequent two intervals. Minor local fluctuations are present in the pre-peak region (a slight decrease from step 1,000 to step 2,000), but do not affect the overall unimodal structure.

A key observation is that D_{nov} remains in a high-value regime (> 0.90) from step 1,000 through step 32,000 in the 160m model, compared to rapid decay beginning at step 4,000 in the 410m model. This extended saturation is consistent with a delayed transition to negative-dominated dynamics, corresponding to a later peak. The mechanism is fully consistent with the Friction Update dynamics derived in Section 2.

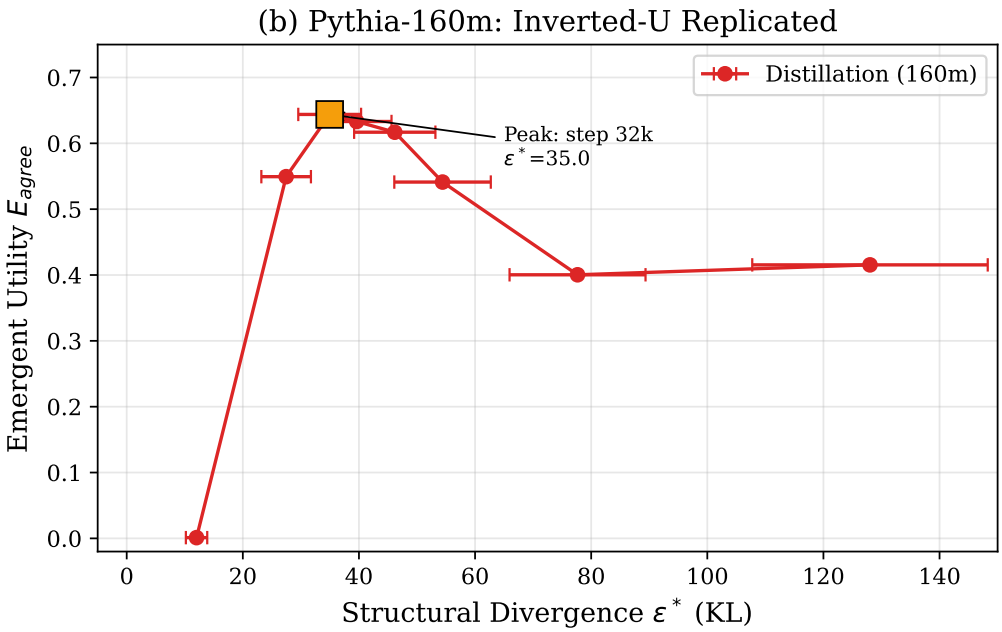

Figure 2: Pythia-160m distillation track: E_{agree} vs. ε^*. The inverted-U is replicated, peaking at step 32,000 ($\varepsilon^* = 34.96$). Step 0 ($\varepsilon^* = 690.2$, $E_{\text{agree}} = 0$) is omitted for readability; see Table 2.

Table 2: Distillation track (Pythia-160m): All measurements. $\star$ marks the peak of E_{agree}.

Step	ε^* (mean $\pm$ SE)	D_{nov}	P_{agree}	E_{agree}
0	690.21 ± 112.35	0.987	0.000	0.000
1,000	128.02 ± 20.28	0.919	0.453	0.415
2,000	77.65 ± 11.71	0.910	0.443	0.400
4,000	54.40 ± 8.31	0.934	0.580	0.541
8,000	46.17 ± 7.00	0.984	0.627	0.617
16,000	39.61 ± 6.00	0.991	0.639	0.633
32,000	34.96 ± 5.41	0.971	0.664	$0.644^\star$
64,000	27.46 ± 4.27	0.904	0.611	0.549
128,000	12.03 ± 1.84	0.002	0.725	0.001

4.3 Cross-Scale Comparison

Table 3 compares the distillation track results across the two model scales.

The unimodal structure of E_{agree} is replicated across model scales under identical experimental conditions, confirming that the phenomenon is not an artifact of a particular model size. However, two scale-dependent differences emerge:

Peak shift. The peak occurs at step 32,000 ($\varepsilon^* = 34.96$) for 160m versus step 8,000 ($\varepsilon^* = 67.57$) for 410m. The present results suggest that $\varepsilon^*_{\text{opt}}$ and the corresponding training step may depend on model scale, though this observation is based on two scales and should be regarded as preliminary.

D_{nov} **saturation range.** The high-D_{nov} regime (> 0.9) persists from step 0 to step 32,000 in the 160m model, compared to step 0 to step 4,000 in the 410m model. This difference in saturation range directly governs the peak shift through the Friction Update dynamics: as long as D_{nov} remains saturated ($\frac{dD_{\text{nov}}}{d\varepsilon^*} \approx 0$), E_{agree} continues to rise with increasing P_{agree}.

Statistical note. For the 160m model, the post-peak trajectory exhibits a strict monotonic decrease across the final two intervals. While the delayed peak structurally limits the number

Table 3: Cross-scale comparison of distillation track results.

	Pythia-160m	Pythia-410m
Peak step	32,000	8,000
ε^* at peak	34.96	67.57
E_{agree} at peak	0.644	0.468
ε^* range	12.0–690	5.8–970
Post-peak monotonic decline	Yes	Yes
Post-peak data points	2	4
$D_{\text{nov}} > 0.9$ range	steps 0–32k	steps 0–4k

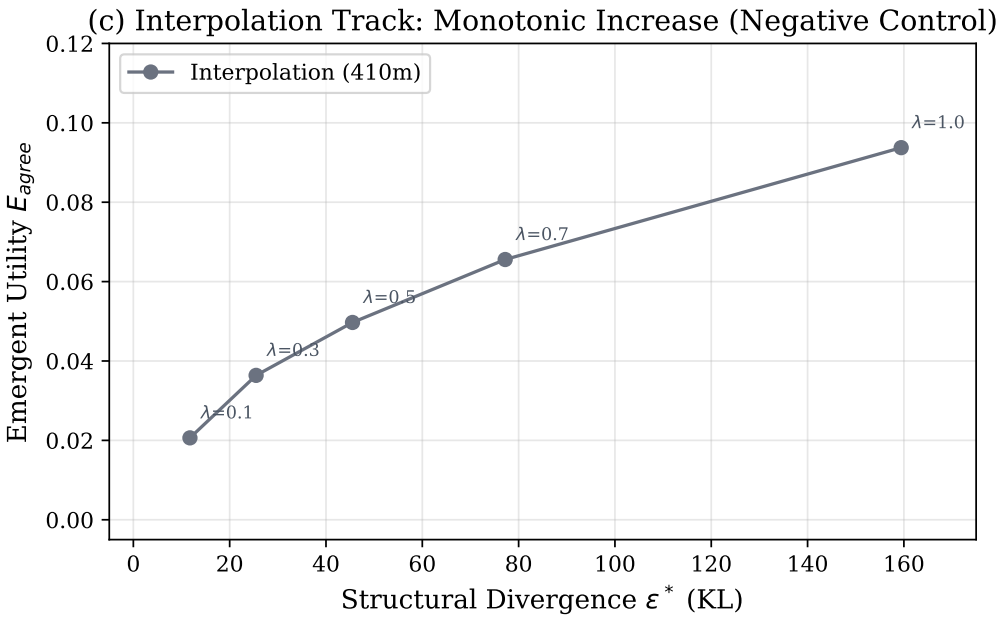

Figure 3: Interpolation track (Pythia-410m): E_{agree} vs. ε^*. Monotonic increase with no peak—negative control confirming process dependence.

of post-peak data points ($n = 3$ including the peak), the observed downward trend is consistent with the 410m results, where four post-peak points yield $r = -0.993$, $p = 0.0007$.

4.4 Interpolation Track: Monotonic Increase (Negative Control)

Figure 3 and Table 4 present measurements from the interpolation track (Track B, Pythia-410m). E_{agree} increases monotonically with ε^*, with no peak observed up to $\varepsilon^* = 159.44$.

This result demonstrates that the emergence of the inverted-U depends on the generative process, supporting hypothesis H2.5 (process dependence).

4.5 D_{nov} Saturation Dynamics: Mechanism of the Inverted-U

The behavior of D_{nov} differs qualitatively between the distillation and interpolation tracks (Figure 4 and Table 5).

In the distillation track, D_{nov} reaches near its upper bound (0.982) at step 0 ($\varepsilon^* = 969.9$) and decays as learning progresses, exhibiting early saturation. In the interpolation track, $D_{\text{nov}} \propto \varepsilon^*$ holds as a linear relationship, with no saturation observed.

As derived in the Theory section, when D_{nov} saturates ($\frac{dD_{\text{nov}}}{d\varepsilon^*} \approx 0$), the negative second term of the dynamical equation dominates, forming the right-side decay of the inverted-U. When

Table 4: Interpolation track (Pythia-410m): Selected measurements.

λ	ε^*	D_{nov}	P_{agree}	E_{agree}
0.10	11.75	0.025	0.838	0.021
0.30	25.49	0.048	0.762	0.036
0.50	45.51	0.072	0.695	0.050
0.70	77.21	0.105	0.623	0.066
1.00	159.44	0.190	0.494	0.094

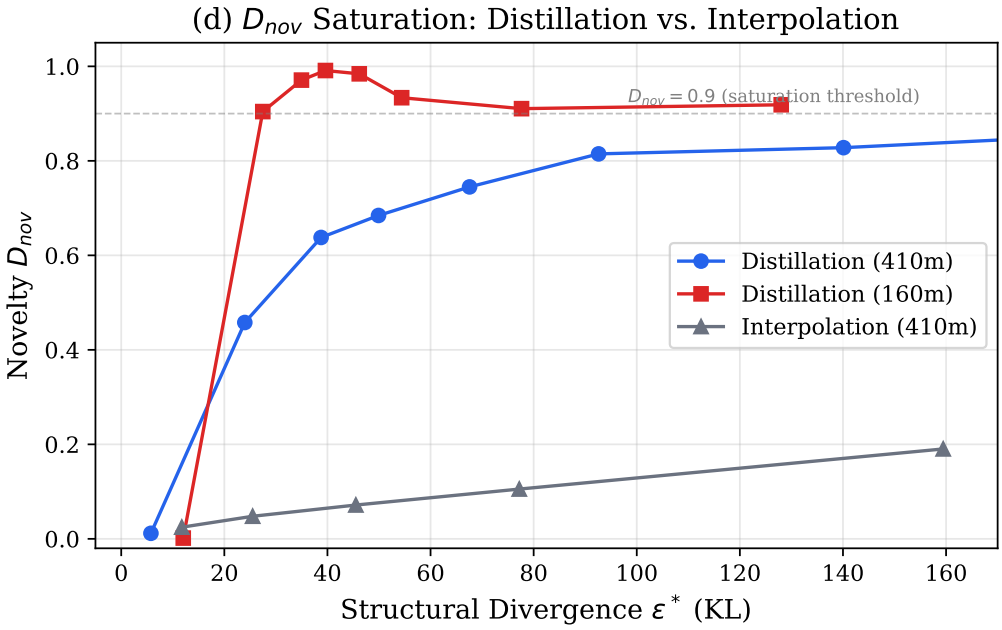

Figure 4: D_{nov} vs. ε^*: distillation (410m, 160m) vs. interpolation (410m). Distillation exhibits early saturation ($D_{\text{nov}} > 0.9$), while interpolation shows linear growth without saturation. The dashed line marks $D_{\text{nov}} = 0.9$.

D_{nov} does not saturate (interpolation), the positive first term consistently prevails, maintaining monotonic increase.

4.6 Model Comparison

Table 6 presents the AIC comparison under both fitting conditions for the 410m data (steps 1,000–128,000, step 0 excluded).

Under weighted fitting, Proposition 1 gamma (3p) achieves the best AIC; under unweighted fitting, LogPeak (3p) achieves the best AIC. The model ranking thus depends on the fitting method. However, under both conditions, all unimodal asymmetric models decisively reject quadratic and linear models (ΔAIC > 20 under unweighted conditions), confirming that the inverted-U structure is robust to model selection.

For the 160m data, LogPeak (3p) achieves the best AIC under both fitting conditions ($R^2 = 0.93$ unweighted, $R^2 = 0.99$ weighted), with Proposition 1 models showing ΔAIC > 11. This indicates that the AIC ranking of specific functional forms is not stable across model scales, reinforcing the position that Proposition 1 should be regarded as a theoretically grounded minimal model rather than a universally optimal functional form.

Table 5: Comparison of D_{nov} at matched ε^* regions (distillation vs. interpolation, 410m).

ε^*	D_{nov} (distill.)	D_{nov} (interp.)	Ratio
~ 7	0.012	0.014	0.8×
~ 25	0.458	0.048	9.6×
~ 45	0.684	0.072	9.6×
~ 77	0.745	0.105	7.1×

Table 6: AIC model comparison (Pythia-410m). ΔAIC is computed from the best model within each condition.

	Weighted (inv. SE)		Unweighted	
Model	AIC	ΔAIC	AIC	ΔAIC
Prop. 1 gamma (3p)	16.12	0.00	−41.17	11.64
Prop. 1 orig (2p)	32.73	16.61	−41.84	10.96
LogPeak (3p)	28.71	12.59	−52.81	0.00
Quadratic (3p)	27.37	11.25	−31.35	21.45
Linear (2p)	33.52	17.40	−27.21	25.59

4.7 Stability of α

The estimated α in the generalized Proposition 1 gamma shows sensitivity to both the fitting method and model scale (Table 7).

Under unweighted fitting for the 410m data, α is stable at 0.81 ± 0.03 across all six robustness conditions (sample sizes $n = 25, 50, 100$; sequence lengths $T = 64, 128, 256$). The AIC difference between Proposition 1 original ($\alpha = 1$ fixed, 2p) and Proposition 1 gamma (α free, 3p) is $\Delta\text{AIC} = 0.67$ (negligible), indicating that $\alpha = 1$ provides a sufficient approximation at the 410m scale.

However, at the 160m scale, the unweighted estimate yields $\alpha = 1.62$, indicating that α is not stable across model scales. This scale dependence further supports treating Proposition 1 as a useful theoretical framework rather than a universal empirical law.

4.8 Additional Findings

$\varepsilon^*(\lambda) \propto \lambda^2$: In the interpolation track, ε^* follows a quadratic function of λ with high precision (Quadratic $R^2 = 0.9998$, Linear $R^2 = 0.9609$), consistent with the Fisher information matrix approximation [3] $\varepsilon^* \approx \frac{1}{2}\Delta W^T G(\theta)\Delta W$. This result was consistently confirmed across five interpolation experiments with different LoRA configurations. Notably, this quadratic relationship confirms that linear interpolation in parameter space does not correspond to a geodesic on the statistical manifold (which would yield $\varepsilon^* \propto \lambda$), but rather traces a nonlinear path in distribution space.

Effect of training conditions on ε^*: Under identical LoRA settings (rank=64, α=256, 10,000 steps), changing training conditions (learning rate, scheduler, training data size) altered ε^* at $\lambda = 1.0$ by a factor of 6.8 (from 23.35 to 159.44), while the weight norm $\|\Delta W\|$ changed by only 1.15×. Thus, ε^* depends on the direction rather than the magnitude of ΔW.

Domain independence: Evaluating a CODE-trained LoRA model on both CODE and WikiText yielded a scale ratio of 0.9×, indicating that evaluation domain has minimal effect on ε^*.

Seed independence: No significant seed effects were detected in any experiment (ANOVA $p > 0.05$ for all metrics across all conditions).

Table 7: Estimated α under different fitting conditions and model scales.

Fitting condition	α (410m)	α (160m)
Weighted (incl. step 0)	2.34	—
Weighted (excl. step 0)	2.12	$12.50^{\dagger}$
Unweighted (all conditions)	0.81 ± 0.03	1.62

†Degenerate fit ($C \approx 0$); not interpretable.

Table 8: Preregistered hypothesis verdicts.

Hypothesis	Verdict	Evidence
H0 (monotonic)	Rejected	Post-peak $r = -0.993$, $p = 0.0007$ (410m)
H1a (inverted-U exists)	Confirmed	Inverted-U confirmed at two scales
H1b (Prop. 1 form)	Not established	AIC rank depends on method and scale
H2.5 (process dep.)	Confirmed	Interpolation: monotonic; Distillation: inverted-U

4.9 Hypothesis Verdict

The existence of the inverted-U structure (H1a) is confirmed across two model scales. The specific functional form of Proposition 1 (H1b) is not established as universally optimal; its AIC ranking varies with fitting method and model scale.

5 Discussion

This study rejected the conventional hypothesis (H0) that complete agreement ($\varepsilon^* \to 0$) maximizes emergence, and statistically confirmed the inverted-U phenomenon in which emergence is maximized at optimal imperfection ($\varepsilon^* > 0$). The phenomenon is replicated across two model scales, strengthening the case for its generality. This section discusses the dynamical conditions producing this phenomenon and the methodological insights from model selection.

5.1 Saturation Dynamics of D_{nov} and Information-Geometric Interpretation

The conditions for the inverted-U are unified by the differential equation $\frac{dE}{d\varepsilon^*} = \frac{dD_{\text{nov}}}{d\varepsilon^*} \times P_{\text{agree}} + D_{\text{nov}} \times \frac{dP_{\text{agree}}}{d\varepsilon^*}$.

Weight interpolation constitutes linear movement in parameter space. Along this trajectory, representational distance (D_{nov}) increases linearly with distributional distance (ε^*), so D_{nov} does not saturate and $\frac{dD_{\text{nov}}}{d\varepsilon^*} > 0$ holds throughout. Consequently, the positive first-term contribution dominates, and emergence increases monotonically. The proposed mechanism therefore predicts that the inverted-U does not form along linear interpolation paths, which is consistent with our experimental observations.

In contrast, distillation is an optimization-induced trajectory exploring the loss landscape via SGD. Along this trajectory, the representation space rapidly orthogonalizes during early learning, and D_{nov} saturates at its upper bound. This yields $\frac{dD_{\text{nov}}}{d\varepsilon^*} \approx 0$, allowing the negative second term to dominate and forming the right tail (decay region) of the inverted-U.

The cross-scale comparison reveals that the duration of D_{nov} saturation varies with model scale: the high-D_{nov} regime (> 0.9) extends from step 0 to step 32,000 in the 160m model but only to step 4,000 in the 410m model. This difference governs the peak shift through the same dynamical mechanism, providing further evidence consistent with the Friction Update equation.

5.2 Model Selection and Methodological Insights on α

AIC-based model comparison revealed that the optimal functional form changes depending on both the fitting method and the model scale. For the 410m data, weighted least squares selects Proposition 1, while unweighted least squares selects LogPeak. For the 160m data, LogPeak achieves the best AIC under both conditions.

LogPeak provides the highest data-driven fit but lacks theoretical grounding in its parameters. In contrast, Proposition 1 is a minimal parametric model with a theoretical foundation in rate-distortion theory [4, 14]. The optimization parameter α is stable at $\alpha \approx 0.8$ under unweighted conditions for the 410m scale, and $\Delta\text{AIC} = 0.67$ (negligible) when fixed at $\alpha = 1$. However, α varies across scales (0.81 for 410m, 1.62 for 160m), indicating that it is not a universal constant.

Crucially, regardless of whether LogPeak or Proposition 1 is employed, and regardless of model scale, quadratic and linear models are decisively rejected ($\Delta\text{AIC} > 10$). This highlights a separation between descriptive accuracy and theoretical interpretability. The emergence function in this system belongs to the class of "unimodal, asymmetric, right-tail-decaying functions," and the existence of the inverted-U phenomenon is a robust fact independent of model selection or scale.

5.3 Limitations

First, although the inverted-U is confirmed across two model scales (160m and 410m) within the Pythia family, all experiments use a single architecture (GPT-NeoX). Generalization to different architectures (e.g., encoder-decoder models, mixture-of-experts) remains unverified.

Second, the "student models" in the distillation track represent different learning stages of the same model, rather than truly independent pretrained systems. Evaluation of dynamics between independently trained models is a task for future work.

Third, the definition of emergent utility $E = D_{\text{nov}} \times P_{\text{agree}}$ depends on specific operational choices. Whether structural properties are preserved under alternative definitions of D_{nov} (e.g., using intermediate-layer representations) or alternative agreement metrics (e.g., top-k agreement, JSD, cosine similarity between softmax outputs) requires additional verification.

Fourth, extension of this theory to heterogeneous cognitive systems such as human–AI collaboration remains at the level of suggestion and requires empirical confirmation.

Fifth, the 160m distillation track provides only two post-peak data points (three including the peak), limiting the statistical strength of the post-peak decline assessment at this scale. The observed monotonic decrease is consistent with the 410m results, but additional intermediate checkpoints would strengthen the evidence.

Sixth, Theorem 2 relies on the injectivity of the embedding map ϕ with respect to the output distribution p. This is adopted as an architecture-dependent working assumption; because ϕ involves token-wise averaging and normalization, injectivity is not guaranteed in general. The theorem's scope is therefore limited to architectures and conditions under which this assumption holds.

6 Conclusion

This study demonstrated the existence of an inverted-U relationship between structural divergence (ε^*) and emergent utility (E) with statistical significance ($p = 0.0007$ for Pythia-410m). The unimodal structure is replicated in Pythia-160m under identical experimental conditions, supporting the presence of the phenomenon across multiple model scales. This result shows that emergence depends not on the magnitude of divergence, but on the nature of the generative dynamics. The experimental evidence is consistent with generative dynamics playing a causal role in the formation of the inverted-U structure.

The inverted-U appears only under optimization-induced dynamics based on SGD (distillation); under linear weight interpolation in parameter space, it is not observed. The dynamical prerequisite is the early saturation of novelty D_{nov} accompanying SGD-driven exploration, as established through the differential equation analysis. The cross-scale comparison further reveals that the range over which D_{nov} remains saturated varies with model size, governing the position of the optimal divergence $\varepsilon^*_{\mathrm{opt}}$ through the same dynamical mechanism.

While multiple asymmetric functions can approximate the phenomenon in a data-driven manner, Proposition 1 ($E = C\varepsilon^* \exp(-k\varepsilon^*)$) serves as a useful minimal model with theoretical grounding. Its AIC ranking varies with fitting method and model scale, and the phenomenon itself—the inverted-U structure and its dependence on optimization-induced dynamics—is the robust, scale-invariant finding of this work.

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